

**MACHINE LEARNING APPROACHES TO DETECT
And Predict
NEUROCOGNITIVE DISORDERS AMONG OLDER ADULTS IN INDIA:
INSIGHTS FROM LASI-DAD WAVE 1**

By

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Declaration

I hereby declare that the work presented in this dissertation titled "**Machine Learning Approaches to Detect and Predict Neurocognitive Disorders Among Older Adults in India: Insights from LASI-DAD Wave 1**" has been carried out entirely by me, under the guidance of my supervisor Dr. Preeti Dhillon, Assistant Professor, Department of Survey Research and Data Analytics, International Institute for population Sciences, Mumbai for the partial fulfilment of the requirement for the award of the degree Master of Science in Survey Research and Data Analytics at the International Institute for Population Sciences, Mumbai. I further declare that this work is original and has not been submitted in part or in full to any other University/Institute for the award of any Degree/Diploma.

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ABSTRACT

Background: India faces a growing neurocognitive disorder burden MCI prevalence of 17.6% and dementia prevalence of 7.2% among adults aged 60+ yet community-deployable screening tools and mechanistic evidence for modifiable pathways remain lacking.

Objectives: (1) Develop ML models for MCI detection; (2) predict NCD using only non-cognitive, modifiable risk factors; and (3) quantify IADL-mediated attenuation of the nutritional status (MNA)-NCD association.

Methods: Secondary analysis of LASI-DAD Wave 1 (N = 2,528). Zero-leakage ML pipeline with MICE imputation, SMOTE, Youden thresholds, and SHAP explainability. Three nested logistic regression models estimated IADL mediation.

Results: The Random Forest algorithm achieved an Area Under the Receiver Operating Characteristic Curve (AUROC) value of 0.836, indicating a sensitivity of 81.2%. The IQCODE and Blessed Performance metrics emerged as the predominant predictors. Absent any cognitive assessments, the area under the receiver operating characteristic curve (AUROC) was 0.732, indicating a 87% recovery of detection performance. The Instrumental Activities of Daily Living (IADL) scale emerged as the foremost predictor across all four models and all six subgroups. Each one-unit MNA increase was associated with a 9% reduction in NCD odds (OR = 0.91, $p = 0.0006$). The IADL was found to mediate 44.3% of this effect overall, with females demonstrating a 56.1% mediation rate and rural residents showing a 22.5% rate. It has been determined that, at $MNA \approx 8$, the predicted probability of NCD exceeds 80%.

Conclusions: The IADL has been identified as a highly significant modifiable NCD predictor, exhibiting a demonstrable correlation with MNA.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
ADL	Activities of Daily Living
AUROC	Area Under the Receiver Operating Characteristic Curve
CDR	Clinical Dementia Rating
CSID	Community Screening Instrument for Dementia
HMSE	Hindi Mental State Examination
IADL	Instrumental Activities of Daily Living
IQCODE	Informant Questionnaire on Cognitive Decline in the Elderly
LASI-DAD	Longitudinal Ageing Study in India - Diagnostic Assessment of Dementia
LMIC	Low- and Middle-Income Country
MCI	Mild Cognitive Impairment
MICE	Multiple Imputation by Chained Equations
ML	Machine Learning
MNA	Mini Nutritional Assessment
NCD	Neurocognitive Disorder
RF	Random Forest
SHAP	SHapley Additive exPlanations
SMOTE	Synthetic Minority Over-sampling Technique
SVM	Support Vector Machine
TICS	Telephone Interview for Cognitive Status

CHAPTER 1: INTRODUCTION

1.1 Background

Cognitive impairment and dementia constitute one of the most pressing public health challenges of the 21st century. Globally, the number of people living with dementia is projected to increase from 57 million in 2019 to over 152 million by 2050 (WHO, 2023). In India, rapid demographic transition has dramatically amplified this burden. Recent evidence from the Longitudinal Ageing Study in India - Diagnostic Assessment of Dementia (LASI-DAD) indicates a weighted prevalence of mild neurocognitive disorder (MCI) of approximately 17.6% and major neurocognitive disorder (dementia) of 7.2% among adults aged 60 years and above (Kundu et al., 2023). These figures are substantially higher than many earlier Indian estimates and reveal pronounced gradients across age, sex, education, rural-urban residence, and socioeconomic status.

Mild Cognitive Impairment (MCI) signifies a pivotal transitional phase between typical cognitive aging and the onset of dementia. Individuals with MCI demonstrate notable cognitive decline that exceeds expected norms for their age and educational attainment, yet maintain relatively preserved functional independence. This stage presents the optimal opportunity for timely intervention. However, in resource-constrained settings such as India, where access to specialist neurological services remains limited and where fewer than 0.3 neurologists per 100,000 population are available in rural areas, scalable tools for early detection and prevention-oriented risk stratification are urgently required. Projections indicate that India's population aged 60 and above will exceed 340 million by the year 2050 (United Nations, 2022), underscoring the profound national public health implications of this phenomenon.

Modifiable risk factors including vascular conditions, nutritional deficits, functional decline, depression, anxiety, and sensory impairments play a central role and represent actionable targets for policy and clinical practice. Mild Cognitive Impairment represents the critical transitional stage between normal cognitive ageing and dementia. Individuals with MCI exhibit noticeable cognitive decline greater than expected for their age and education level, yet retain relatively preserved functional independence making this the most actionable window for early detection and preventive intervention. However, in resource-constrained settings such as India, where fewer than 0.3 neurologists per 100,000 population are available in rural areas and where formal

cognitive testing infrastructure is largely inaccessible, scalable tools for early detection and prevention-oriented risk stratification are timely required.

1.2 Review of Literature

Global Burden and Epidemiological Context:

Dementia represents one of the most urgent public health crises of the 21st century. Over 55 million people worldwide currently live with dementia, with nearly 10 million new cases diagnosed annually, a trajectory that is expected to accelerate sharply as populations age globally (World Health Organization [WHO], 2023). The condition imposes enormous societal, economic, and caregiving burdens, disproportionately affecting low-and middle-income countries where diagnostic infrastructure remains inadequate. Understanding the determinants of cognitive decline and identifying those amenable to intervention has therefore become a central objective of contemporary gerontological and epidemiological research.

The 2024 Lancet Commission on Dementia Prevention, Intervention and Care represents the most comprehensive synthesis of this evidence to date. The Commission's 2020 report identified 12 modifiable risk factors linked to approximately 40% of all dementia cases. The updated Commission expanded this framework to 14 risk factors, adding uncorrected vision loss and elevated low-density lipoprotein (LDL) cholesterol. Collectively, these factors were estimated to account for up to 45% of global dementia incidence (Livingston et al., 2024). These factors span the life course, including early-life educational quality, midlife exposures such as hypertension, hearing loss, obesity, and traumatic brain injury, and late-life factors including depression, physical inactivity, social isolation, diabetes, smoking, excessive alcohol consumption, and air pollution. This life-course framework has fundamentally reoriented dementia research toward prevention-oriented models and scalable, population-level risk stratification.

Modifiable Risk Factors and Intervention Evidence:

Experimental validation for the modifiability of dementia risk comes primarily from the Finnish Geriatric Intervention Study to Prevent Cognitive Impairment and Disability (FINGER) randomized controlled trial (Ngandu et al., 2015). This landmark multi-domain intervention

comprising dietary guidance, physical exercise, cognitive training, and vascular risk monitoring demonstrated a 25% greater improvement in global cognitive performance in the intervention group compared to controls, with particularly pronounced gains in executive function (83% greater improvement) and psychomotor speed (150% greater improvement). These findings confirm that nutritional and functional pathways to cognitive health are meaningfully modifiable, even in at-risk older adults, and provide theoretical grounding for prevention-oriented prediction tools.

Nutritional status has emerged as a critical and independent modifiable predictor along this pathway. Malnutrition and micronutrient deficiencies in older adults have been associated with sarcopenia, reduced physical activity, functional impairment, and accelerated cognitive decline through inflammatory, vascular, and neurotrophic mechanisms (Solfrizzi et al., 2011; Scarmeas et al., 2018). The utilization of validated instruments, such as the Mini Nutritional Assessment (MNA; Vellas et al., 1999), has enabled the operationalization of nutritional status in large-scale epidemiological surveys. It is now widely accepted that the quality and nutritional content of one's diet can play a significant role in reducing the risk of developing dementia. This reduction in risk is achieved through a variety of interconnected mechanisms, including inflammatory modulation, vascular protection, and the maintenance of synaptic plasticity. These findings underscore the critical importance of nutrition as a primary target for prevention strategies (Scarmeas et al., 2018).

Functional Decline as a Predictor of Dementia

Functional status, particularly as captured by Instrumental Activities of Daily Living (IADL), occupies a central position in the early detection and prediction of cognitive impairment. Marshall et al. (2011) demonstrated that IADL limitations are among the earliest and most sensitive indicators of preclinical dementia, outperforming direct cognitive test scores in functional prediction models. This finding is consistent with clinical observations that subtle difficulties in managing finances, medications, and complex tasks often precede formal cognitive diagnoses by several years. Despite this, few population-level studies have formally quantified the extent to which nutritional status influences cognitive outcomes *through* functional decline the instrumental activities of daily living (IADL)-mediated pathway leaving an important mechanistic and methodological gap in the literature.

Risk factor-based predictive models that incorporate vascular, functional, and lifestyle variables have achieved AUROC values of 0.68-0.74 in community samples, establishing a useful benchmark for non-cognitive prediction paradigms (Stephan et al., 2010). These models demonstrate that clinically actionable predictions are achievable without relying on extensive neuropsychological test batteries, a particularly important consideration for screening in resource-limited settings.

Machine Learning Approaches to Cognitive Impairment Detection

The application of machine learning (ML) to cognitive impairment classification has expanded considerably over the past decade, with ensemble methods consistently outperforming traditional statistical approaches. A systematic review by Ansart et al. (2021, *Medical Image Analysis*) established that Random Forest and gradient boosting (XGBoost) ensembles achieve AUROC values of 0.80-0.90 for mild cognitive impairment (MCI) and dementia detection in population-level survey data, surpassing the performance of logistic regression and Support Vector Machine (SVM) classifiers. These ensemble architectures capitalize on non-linear feature interactions, particularly between vascular, nutritional, and functional variables, that linear models fail to capture.

Explainability remains a critical concern in clinical ML deployment. SHAP (SHapley Additive exPlanations), developed by Lundberg and Lee (2017) on game-theoretic principles, has become the de facto standard for feature attribution in clinical prediction models. SHAP values provide individualized, directional explanations of model predictions, enabling clinicians and policymakers to translate complex model outputs into interpretable clinical narratives a prerequisite for real-world adoption in healthcare settings.

1.3 Research Gap

Despite the growing body of evidence on cognitive impairment risk and machine learning-based prediction, significant data, methodological, and analytical gaps persist particularly in the Indian context. The Longitudinal Ageing Study in India - Diagnostic Assessment of Dementia (LASI-DAD) Wave 1 remains India's most comprehensive, nationally representative cognitive health dataset, yet it has been substantially underutilized in ML-based neurocognitive disease (NCD) prediction research, critically limiting the population-level generalizability of existing findings.

A more fundamental methodological constraint is that the majority of ML dementia classification models including those applied to LASI-DAD are built on cognitive test scores such as the HMSE, IQCODE, and CSID as primary predictors. While such models achieve high discriminative performance, they are inherently unsuitable for community-level or primary-care screening in India's low-resource, low-literacy, and linguistically diverse settings where formal cognitive testing infrastructure is largely unavailable. This creates a critical disconnect between diagnostic model development and the real-world prevention-oriented applications that India's public health system most urgently requires.

At the mechanistic level, no study to date has formally quantified the extent to which nutritional status influences cognitive outcomes through functional decline specifically, the IADL-mediated pathway within a nationally representative Indian sample. This leaves a foundational question in the nutrition-cognition relationship unanswered and precludes the design of targeted, pathway-specific interventions for Indian older adults. Compounding this gap, existing ML studies on mild cognitive impairment /mild dementia risk in India rarely stratify analyses by rural versus urban residence, educational attainment, or sex. This methodological omission conceals critical heterogeneity in risk profiles and predictor importance across sociodemographic subgroups heterogeneity that is essential for designing equity-sensitive screening policies and targeted prevention programs that reflect India's deeply stratified social and geographic landscape. Collectively, these gaps in data utilization, predictor selection, mechanistic understanding, and subgroup analysis define the core intellectual space that the present dissertation seeks to address.

1.4 Research Questions

1. Can machine learning models accurately detect Mild Cognitive Impairment ($CDR = 0.5$) using a combination of cognitive, functional, and demographic predictors, and which predictors drive model performance?
2. Can cognitive impairment ($CDR \geq 0.5$) be predicted with clinically useful performance using only non-cognitive and modifiable risk factors practical for low-resource primary-care and community settings, and does model performance and predictor importance vary across rural/urban, education, and sex subgroups?

3. To what extent does functional decline (IADL) mediate the association between nutritional status (MNA score) and cognitive impairment, and does this mediation differ by sex, education, and rural/urban residence?

1.5 Research Objectives

Objective 1: To train Machine to detect and predict Mild Cognitive Impairment.

Objective 2: To know the possibility of developing tool for low resource settings to detect Dementia Without Clinical Tests using Machine learning.

Objective 3: To examine Nutritional Pathways to Dementia at early stage.

1.6 Structure of the Dissertation

Chapter 1 Introduction

Chapter 2 Data and Methodology.

Chapter 3 Machine Learning Analysis for Mild Cognitive Impairment Detection

Chapter 4 Prevention-oriented Prediction of Cognitive Impairment Using Non cognitive and Modifiable Factors

Chapter 5 Nutritional Status, Functional Decline, and Cognitive Impairment: A Mechanistic Analysis

Chapter 6 Discussion and Conclusion

CHAPTER 2: DATA AND METHODOLOGY

2.1 LASI-DAD: The Data Source

LASI-DAD Wave 1 is the most comprehensive cognitive and functional assessment of Indian older adults conducted to date. Embedded within the nationally representative LASI study, LASI-DAD administered a dedicated dementia diagnostic module to individuals aged 60 and above across all major Indian states. The diagnostic battery included harmonised cognitive tests (HMSE, TICS-m, CSID), informant assessments (IQCODE, Blessed Performance and Habits), functional measures (ADL, IADL, mobility), clinical assessments (blood pressure, diabetes, sensory impairment), psychological scales (depression, anxiety, stress), and the Mini Nutritional Assessment. CDR classification was performed using the algorithm-driven harmonised protocol (McArdle et al., 2023). The breadth of this assessment makes LASI-DAD uniquely suited for the multi-objective analysis undertaken here.

2.2 Study Design and Data Source

This dissertation is a secondary analysis of cross-sectional data from Wave 1 of LASI-DAD, collected between 2017 and 2019 through face-to-face interviews, cognitive testing, informant reports, and clinical assessments. The multistage stratified cluster sampling design ensures national representativeness across urban and rural areas and all major Indian states. The publicly available harmonised dataset (H_LASI_DAD_b1.dta) was used. Initial data preparation, variable creation, and quality checks were performed in Stata; all analytical work was conducted in Python 3.

2.3 Sample Selection

The initial sample consisted of all 2,528 participants who completed the diagnostic assessment and had a valid CDR score. The analytic sample was restricted to individuals aged 60 years and above with non-missing CDR data. Missing values on predictor variables were handled through MICE as described below. Analytic samples varied by chapter objective:

2.4 Outcome Variables

Cognitive status was operationalised using the Clinical Dementia Rating (CDR) scale, a globally validated instrument integrating cognitive performance and functional status through participant and informant interviews. Two binary outcome definitions were used: (i) Chapter 3 mci_vs_normal: CDR = 0 versus CDR = 0.5 (MCI prevalence 67.1%); (ii) Chapters 4 and 5 cdr_impaired: CDR = 0 versus CDR $\geq$ 0.5 (NCD prevalence 69.6%).

2.5 Predictor Variables

Predictors were selected based on theoretical relevance, established epidemiological evidence, modifiability, and practical availability in low-resource settings. A key design distinction separated the Chapter 3 predictor set (which includes cognitive performance variables) from the Chapter 4 and 5 predictor sets (which deliberately exclude all cognitive variables to prevent target leakage).

Cognitive and Informant-Based Measures (Chapter 3 only):

HMSE, TICS, CSID, Judgment score, IQCODE, and Blessed Performance score.
Excluded from Chapters 4 and 5 as direct components of or proxies for the CDR assessment protocol.

Sociodemographic:

Age (years), education (years), sex (female, binary), literacy (binary), rural (rural=0, urban=1), marital status (binary).

Clinical/Vascular:

Systolic blood pressure (mmHg), diabetes (binary), hearing_vision impairment (binary).

Functional Status:

ADL (0-6), IADL (0-7), mobility score. IADL serves as predictor in Chapters 3 and 4, and as proposed mediator in Chapter 5.

Psychological:

Depression score, anxiety score, perceived stress (binary).

Nutritional Status (Chapters 4 and 5):

Mini Nutritional Assessment (MNA) score (continuous; higher = better nutrition).
Primary exposure in Chapter 5.

2.6 Statistical and Machine Learning Methods

2.6.1 General Pipeline Principles

All three analytical chapters followed a common zero-leakage pipeline: Split ->ImbPipeline (Imputer ->SMOTE ->[Scaler for SVM] ->Model) ->Stratified 5-Fold CV ->Held-Out Test Evaluation ->Youden Threshold ->SHAP. A fixed random seed (42) was used throughout all stochastic steps.

2.6.2 Data Splitting and Preprocessing

Data were partitioned into training (80%) and test (20%) sets using stratified random sampling (sklearn StratifiedKFold). All preprocessing MICE imputation, StandardScaler normalisation, and correlation pruning were applied using parameters derived exclusively from training data and applied as transforms to test data. Correlation pruning was conducted on training data only, using a $|r| = 0.70$ threshold with domain-knowledge-based retention decisions.

2.6.3 Missing Value Imputation

Missing values were handled using MICE (sklearn IterativeImputer with BayesianRidge estimator, 10 iterations) embedded as the first pipeline step for Chapters 3 and 4 ensuring per-fold fitting during cross-validation and transform-only application to test data. For Chapter 5 (logistic regression), MICE was applied once to the full dataset following standard inferential practice. Diabetes had the highest missingness (31.58-31.65% in training data), reflecting a structural LASI-DAD data collection gap. All other variables had missingness below 4.5%.

2.6.4 Class Imbalance and SMOTE

Class imbalance (MCI prevalence 67.1%; NCD prevalence 69.6%) was addressed using SMOTE (imbalanced-learn), embedded inside ImbPipeline after imputation. Placing SMOTE inside the pipeline not outside ensures synthetic samples are generated afresh per fold and never contaminate the test set, preventing the data leakage that inflates performance estimates in many published ML clinical studies.

2.6.5 Classifiers

Four supervised classifiers were trained: Random Forest (n_estimators=300, max_depth=8), Gradient Boosting (n_estimators=200, max_depth=5), XGBoost (n_estimators=300, max_depth=6), and SVM (RBF kernel, C=1.0). Tree-based models used imputed but unscaled data; SVM used StandardScaler within the pipeline. Hyperparameters were fixed at stated values for reproducibility.

2.6.6 Evaluation Metrics and Threshold Selection

Six metrics were reported: AUROC (primary), Accuracy, Sensitivity, Specificity, F1-score, and Cohen's Kappa. Decision thresholds were determined by Youden's J index (argmax [sensitivity + specificity - 1]) derived from out-of-fold cross-validation probabilities and applied consistently to the test set, replacing the clinically inappropriate default 0.5 cutoff.

2.6.7 SHAP Explainability

SHAP values were computed for the best-performing model per chapter using TreeExplainer (Random Forest, tree-based models) or KernelExplainer (SVM). SHAP was applied to test-set data, extracting values. Feature importance was summarised through mean |SHAP| bar plots and beeswarm plots.

2.6.8 Mediation Analysis (Chapter 5)

Three nested logistic regression models were estimated using statsmodels Logit on the fully imputed sample (N = 2,528): Model 1 (MNA only), Model 2 (MNA + covariates), and Model 3 (Model 2 + IADL mediator). The IADL-mediated proportion of the MNA-NCD association was estimated as:

$$\text{Attenuation (\%)} = \frac{(\beta_{\text{Model 2}} - \beta_{\text{Model 3}})}{|\beta_{\text{Model 2}}|} \times 100$$

Model fit was assessed using McFadden's pseudo-R². Dose-response patterns were examined via MNA tertiles, and subgroup-specific mediation was estimated within rural, urban, no-education, male, and female strata.

2.7 Software and Reproducibility

All ML analyses were implemented in Python 3.10 using: pandas, numpy, scikit-learn, imbalanced-learn, xgboost, shap, statsmodels, matplotlib, and seaborn and others. Data preparation was conducted in Stata/SE 17. All code is available upon request.

2.8 Ethical Considerations

This study is a secondary analysis of de-identified, publicly available LASI-DAD Wave 1 data. No additional ethical approval was required. The original LASI-DAD study received clearance from the University of Southern California IRB, IIPS, and relevant Indian ethics bodies. All analyses were conducted in accordance with responsible secondary data use guidelines.

CHAPTER 3: Machine Learning Analysis for Mild Cognitive Impairment Detection

3.1 Overview

This chapter presents results from the ML analysis to detect Mild Cognitive Impairment (MCI; CDR = 0.5) against normal cognition (CDR = 0) in LASI-DAD Wave 1. Nineteen clinically relevant predictors spanning cognitive tests, functional status, and sociodemographic variables were used. The analysis employed a publication-grade, leakage-free pipeline (stratified split, MICE within pipeline, SMOTE per fold, Youden threshold, SHAP).

3.2 Analytic Sample

The final analytic sample comprised 2,336 participants aged 60 and above who were present in wave 1 for clinical dementia testing: 768 (32.9%) classified as normal cognition (CDR = 0) and 1,568 (67.1%) as MCI (CDR = 0.5) as provided by harmonized lasi-dad dataset. Data were split into training (n = 1,868) and test (n = 468) sets using stratified random sampling.

3.3 Descriptive Statistics

Table 3.1 summarises baseline characteristics stratified by MCI status. MCI participants were older (69.6 vs. 69.5 years), had lower education (3.2 vs. 4.3 years), higher IADL difficulty (2.5 vs. 1.9), elevated depression (10.5 vs. 9.1) and anxiety (3.3 vs. 2.3), higher IQCODE (3.6 vs. 3.4), and substantially higher Blessed Performance scores (1.4 vs. 0.3) a lower score is better as it indicates normal functioning. Rural prevalence was higher among MCI participants (70.7% vs. 29.3%).

Table 3.1: Descriptive Statistics by MCI Status (N = 2,336, Training Data)

Variable	Overall (N=2,336)	No Impairment (N=768)	MCI (N=1,568)
age (years)	69.1 (7.2)	68.1 (6.4)	69.6 (7.5)
education (years)	3.8 (4.7)	5.0 (4.9)	3.2 (4.4)
systolic_bp (mmHg)	137.9 (23.1)	137.7 (22.0)	138.0 (23.6)
adl	1.2 (1.7)	0.8 (1.3)	1.4 (1.8)
iadl	2.0 (2.1)	1.1 (1.6)	2.5 (2.2)

Variable	Overall (N=2,336)	No Impairment (N=768)	MCI (N=1,568)
mobility	1.9 (0.3)	2.0 (0.2)	1.9 (0.3)
depression	9.7 (5.2)	8.1 (4.9)	10.5 (5.1)
anxiety	2.9 (3.3)	1.9 (2.6)	3.3 (3.5)
iqcode	3.4 (0.4)	3.1 (0.3)	3.6 (0.4)
blessed_perf	1.1 (1.4)	0.3 (0.6)	1.4 (1.6)
Female, n (%)	1208 (51.7%)	367 (47.8%)	841 (53.6%)
Male, n (%)	1128 (48.3%)	401 (52.2%)	727 (46.4%)
Literate, n (%)	1134 (48.5%)	275 (35.8%)	859 (54.8%)
Illiterate, n (%)	1202 (51.5%)	493 (64.2%)	709 (45.2%)
Rural, n (%)	1513 (64.8%)	404 (52.6%)	1109 (70.7%)
Urban, n (%)	823 (35.2%)	364 (47.4%)	459 (29.3%)
Married, n (%)	1586 (67.9%)	550 (71.6%)	1036 (66.1%)
Unmarried, n (%)	750 (32.1%)	218 (28.4%)	532 (33.9%)
Diabetes, n (%)	318 (13.6%)	127 (16.5%)	191 (12.2%)
No Diabetes, n (%)	1298 (55.6%)	393 (51.2%)	905 (57.7%)
Hearing/Vision Impairment, n (%)	1268 (54.3%)	346 (45.1%)	922 (58.8%)
No Hearing/Vision Impairment, n (%)	1061 (45.4%)	416 (54.2%)	645 (41.1%)

Note: Values are mean (SD) for continuous variables; n (%) for categorical variables. MCI = CDR = 0.5.

3.4 Missing Value Analysis

Missingness was low across all predictors (maximum 2.42%). The highest missing rates were diabetes, systolic_bp (2.42%) and blessed_perf (2.08%). Six variables had zero missingness: age, education, female, rural, HMSE, TICS, CSID, and IQCODE.

3.5 Multicollinearity Analysis

Correlation analysis in training data identified one pair exceeding $|r| = 0.55$. Both variables were retained based on their distinct conceptual contributions in the Indian LMIC context. Figure 3:0 displays the full correlation heatmap.

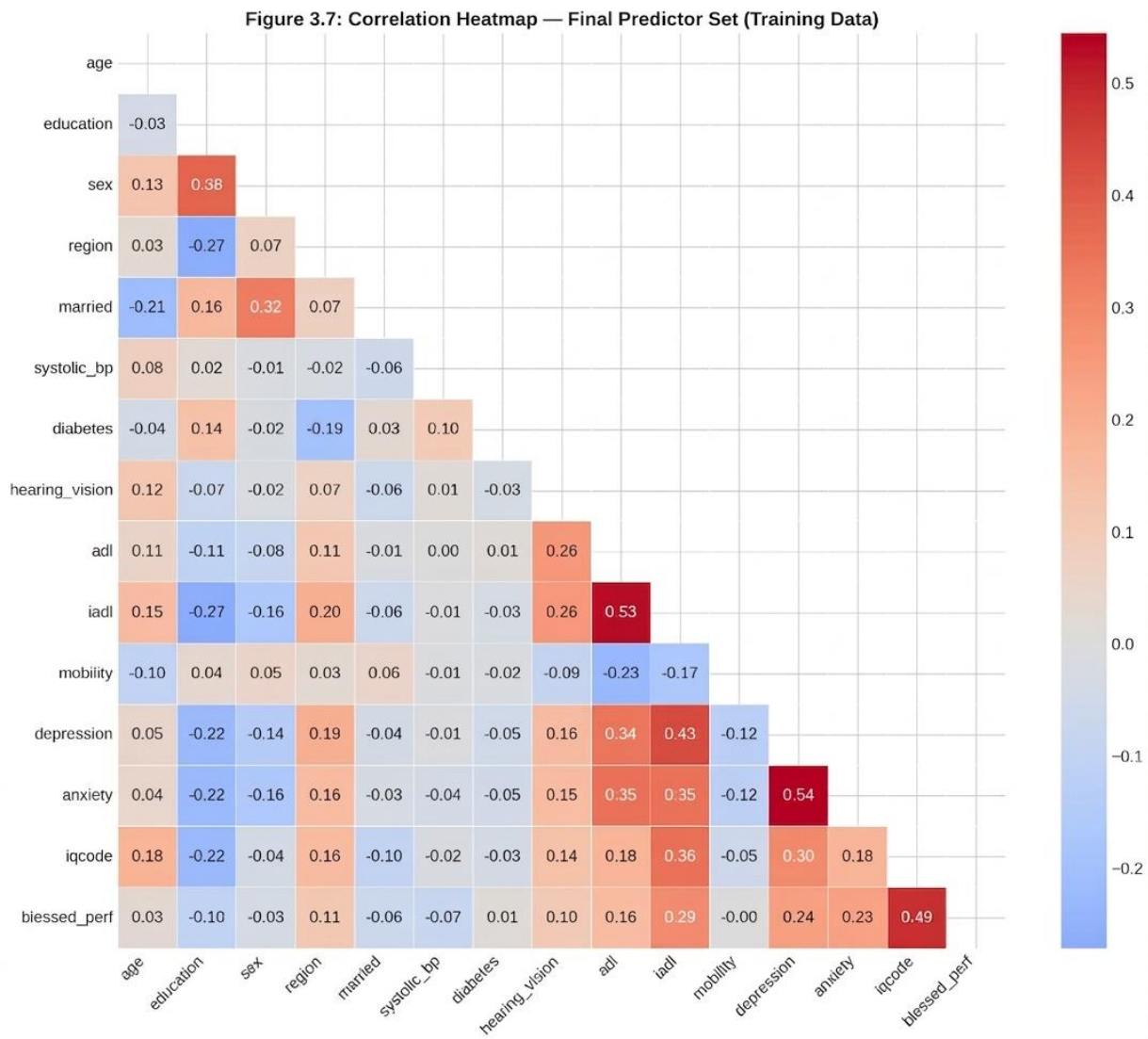


Figure 3.0: Correlation Heatmap of Predictors (Training Data). All other pairs show acceptable orthogonality.

3.6 Model Performance

Table 3.2 presents test-set performance of all four classifiers at Youden-optimal thresholds. Random Forest achieved the highest AUROC (0.836) and sensitivity (81.2%), making it the optimal model for screening. SVM achieved the highest specificity (73.4%) and Kappa (0.498), making it preferable when balanced classification is prioritised. XGBoost achieved the highest sensitivity (82.2%) but with a lower AUROC (0.779). Gradient Boosting achieved AUROC 0.832 with balanced sensitivity/specificity at threshold 0.573 among all these models from different families using different gradient algorithm for learning best patterns.

Table 3.2: Test-Set Performance of ML Classifiers for MCI Detection (n = 468)

Model	AUROC	Accuracy	Sensitivity	Specificity	F1	Kappa	Threshold
Random Forest	0.836	0.786	0.812	0.734	0.836	0.530	0.467
Gradient Boosting	0.835	0.778	0.771	0.792	0.823	0.528	0.573
XGBoost	0.797	0.731	0.822	0.583	0.791	0.415	0.930
SVM	0.824	0.759	0.774	0.734	0.799	0.498	0.623

Note: Threshold = Youden's J optimal cutpoint. AUROC = primary metric. RF achieves best AUROC and sensitivity; SVM achieves best Kappa and specificity.

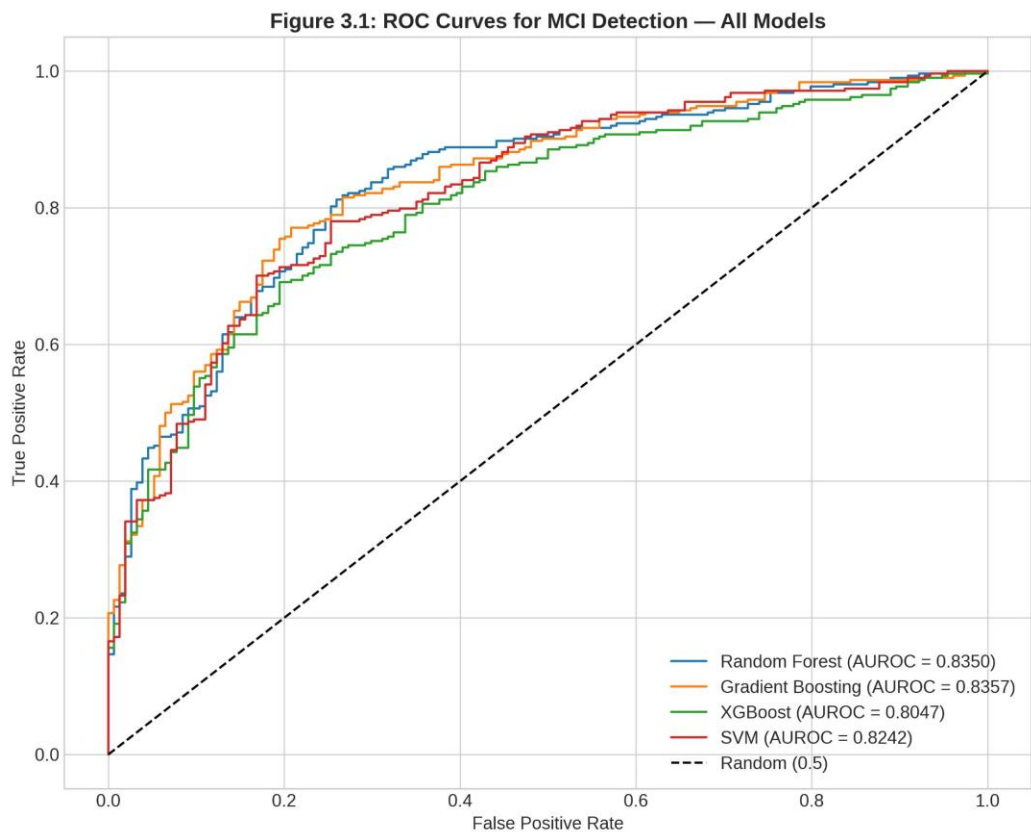


Figure 3.1: ROC Curves All Models for MCI Detection. Random Forest (AUROC=0.836) and SVM (AUROC=0.824) dominate across all thresholds. Dashed line = chance (0.50).

Figure 3.2: Confusion Matrix — Gradient Boosting (Threshold = 0.573)

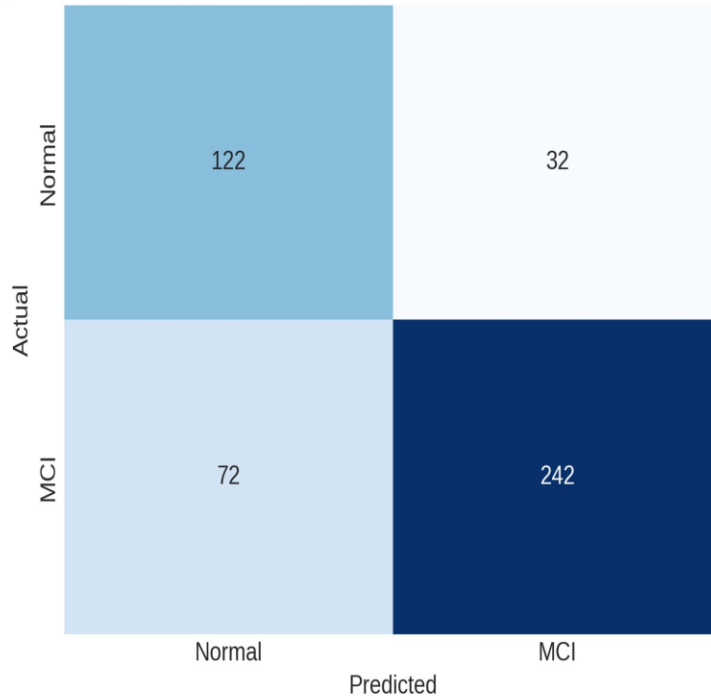


Figure 3.2: Confusion Matrix Random Forest (Threshold=0.467). TP=255, TN=113, FP=41, FN=59. Sensitivity=81.2%; Specificity=73.4%.

3.7 SHAP Feature Importance

SHAP values computed for Random Forest confirm IQCODE as the strongest predictor (mean $|\text{SHAP}| \approx 0.138$) consistent with Jorm (2004), whose systematic review established informant-reported cognitive decline as among the most reliable early dementia indicators, particularly in low-literacy populations, followed closely by Blessed Performance (≈ 0.108). These two informant-based measures together account for the dominant share of predictive signal consistent with their role in capturing informant-perceived cognitive change and day-to-day functional memory failures. IADL ranked third (≈ 0.052) consistent with Marshall et al. (2012), who demonstrated that IADL limitations are among the earliest and most sensitive predictors of preclinical dementia rural residence fourth (≈ 0.034) reflecting the well-documented rural NCD disadvantage in India (Shaji et al., 2010; Kundu et al., 2023) and hearing/vision impairment fifth (≈ 0.022) both sensory impairment and rural residence have been identified as modifiable dementia risk factors in the 2024 Lancet Commission framework (Livingston et al., 2024). Depression, diabetes, ADL, and anxiety formed a secondary contributing tier, consistent with all four being among the 14 modifiable dementia risk factors identified by Livingston et al. (2024).

Stress and mobility contributed negligible independent information. The beeswarm plot confirms that higher IQCODE values (greater informant-reported decline) and higher Blessed Performance scores consistently increase predicted MCI probability, while lower HMSE scores are uniformly risk-increasing.

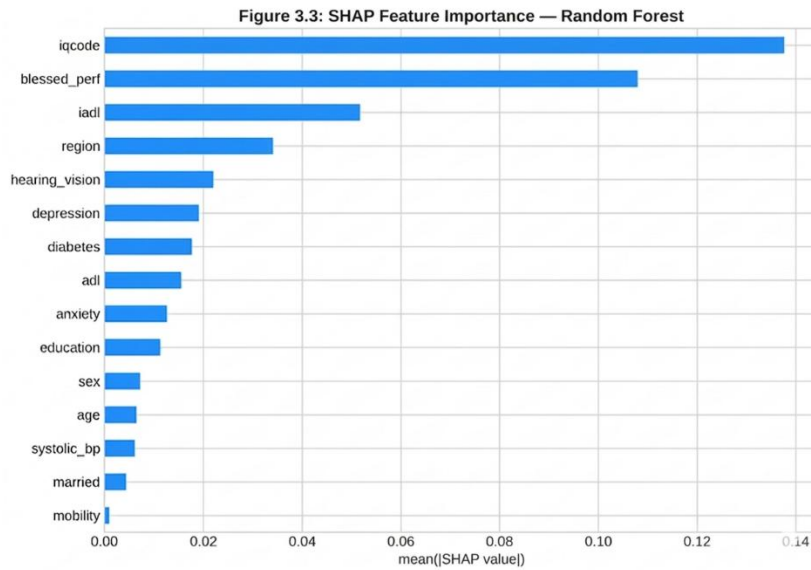


Figure 3.3: SHAP Feature Importance Bar Plot Random Forest. IQCODE and Blessed Performance dominate; all other predictors contribute substantially smaller increments.

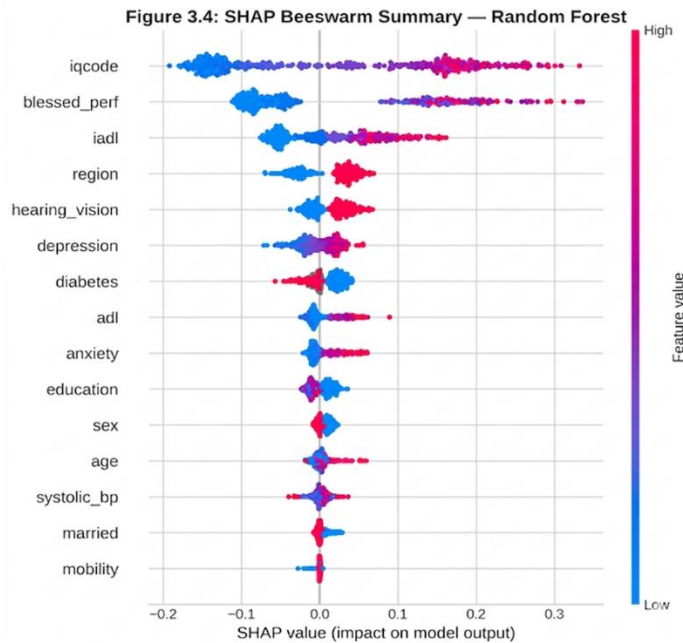


Figure 3.4: SHAP Beeswarm Plot Random Forest. Higher IQCODE (red, rightward) consistently increases predicted MCI probability. The beeswarm confirms logical, clinically coherent directional effects across all predictors.

Chapter 4: Prevention-oriented Prediction of Cognitive Impairment Using Non cognitive and Modifiable Factors

4.1 Overview

This chapter presents results from the prevention-oriented ML framework for predicting NCD ($CDR \geq 0.5$) using exclusively non-cognitive, modifiable, and community-measurable predictors. All cognitive test scores were excluded to ensure deployability in settings without cognitive testing infrastructure. The chapter additionally presents subgroup heterogeneity analysis across six strata: rural/urban, education level, and sex.

4.2 Analytic Sample and Outcome

The analytic sample comprised 2,528 participants: 768 unimpaired ($CDR = 0$, 30.4%) and 1,760 cognitively impaired ($CDR \geq 0.5$, 69.6%). Data were split into training ($n = 2,022$) and test ($n = 506$) sets using stratified random sampling. Sixteen non-cognitive predictors were included: age, education, female, literacy, rural, married, systolic_bp, diabetes(self-reported), hearing&vision, ADL, IADL, mobility, depression, anxiety, stress, and MNA.

4.3 Descriptive Statistics

Table 4.1 presents baseline characteristics. Key group differences: impaired participants were older (70.2 vs. 68.1 years, gap = + 2.1 years), had less education (3.0 vs. 5.0 years, gap = -2.0 years), substantially higher IADL difficulty (2.8 vs. 1.1, gap = +1.7), worse nutritional status (MNA 9.9 vs. 10.8), higher depression (10.8 vs. 8.1) and anxiety (3.4 vs. 1.9), and higher rural prevalence (71.1% vs. 52.6%, gap = 18.5 percentage points). Hearing/vision impairment was more prevalent among impaired participants (59.8% vs. 45.1%).

Table 4.1: Descriptive Statistics by Cognitive Impairment Status (N = 2,528)

Variable	Overall (N=2,528)	No Impairment (N=768)	Cognitive Impairment (N=1,760)
Age (years)	69.6 (7.5)	68.1 (6.4)	70.2 (7.8)
Education (years)	3.6 (4.6)	5.0 (4.9)	3.0 (4.3)
IADL score	2.3 (2.2)	1.1 (1.6)	2.8 (2.3)
ADL score	1.3 (1.8)	0.8 (1.3)	1.6 (1.9)
Mobility score	1.9 (0.3)	2.0 (0.2)	1.9 (0.4)
Depression score	9.9 (5.3)	8.1 (4.9)	10.8 (5.2)
Anxiety score	3.0 (3.3)	1.9 (2.6)	3.4 (3.5)
MNA score	10.2 (2.0)	10.8 (1.9)	10.0 (2.0)
Systolic BP (mmHg)	138.4 (23.4)	137.7 (22.0)	138.7 (23.9)
Rural, n (%)	1,655 (65.5%)	404 (52.6%)	1,251 (71.1%)
Urban, n (%)	873(34.5%)	364(47.4%)	509(28.9%)
Female, n (%)	1,326 (52.5%)	367 (47.8%)	959 (54.5%)
Male, n (%)	1,202 (47.5%)	401 (52.2%)	801 (45.5%)
Hearing/Vision, n (%)	1,399 (55.3%)	346 (45.1%)	1,053 (59.8%)
Diabetes, n (%)	333 (13.2%) *	127 (16.5%)	206 (11.7%)

Note: * Diabetes(self-reported) has 31.65% missingness (structural LASI-DAD gap); interpret with caution.

4.4 Missing Values and Multicollinearity

Diabetes had the highest missingness (31.65%), followed by MNA (4.01%), systolic_bp (2.77%), stress (1.63%), anxiety (1.43%), and depression (1.24%). Six variables had zero missingness: age, education, female, literacy, rural, married. No correlation pair exceeded $|r| = 0.55$.

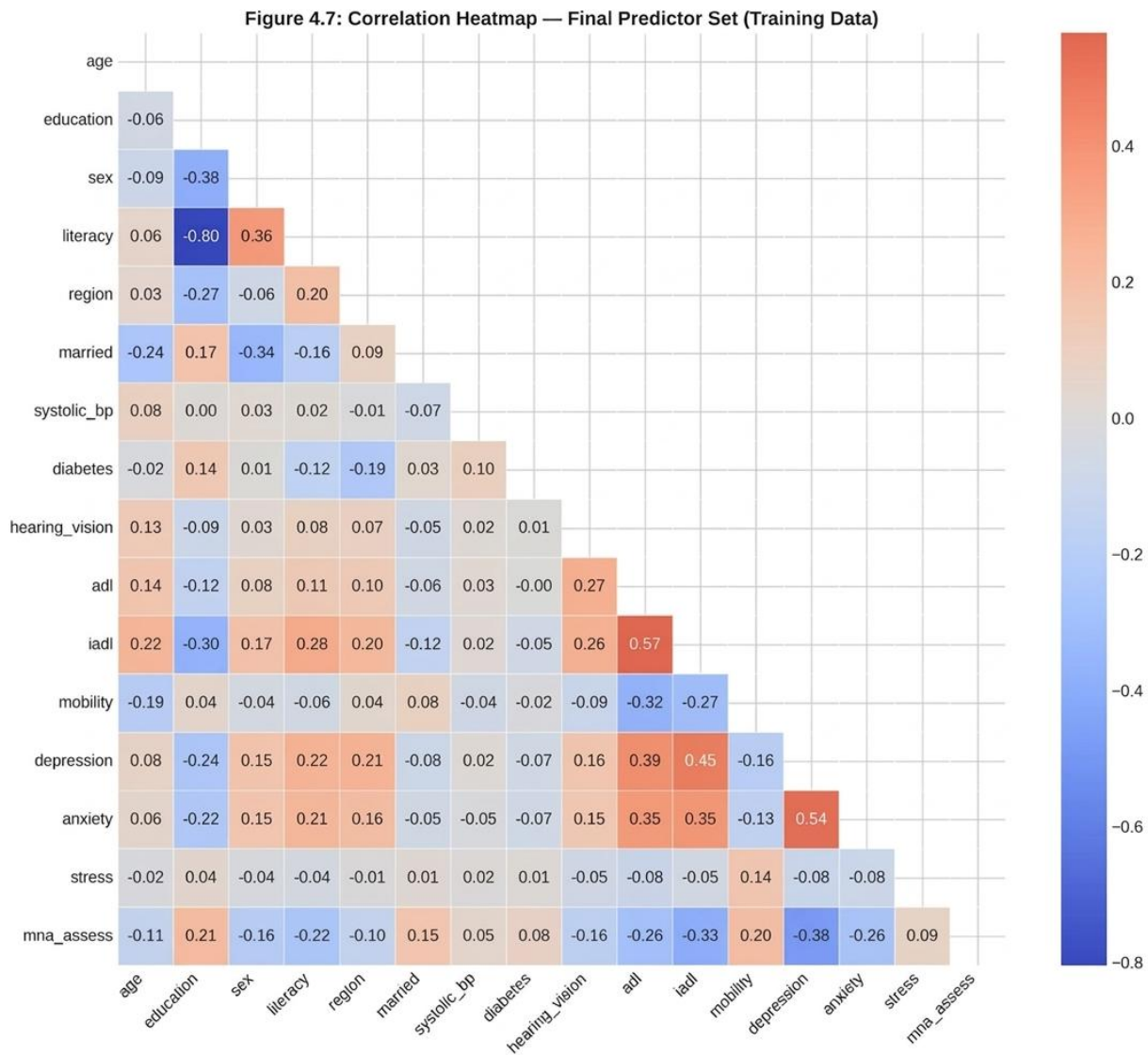


Figure 4.0: Correlation Heatmap Final Predictor Set (Training Data). Highest remaining correlations: depression-anxiety (0.54), ADL-IADL (0.57).

4.5 Model Performance

Table 4.2 presents test-set performance. Random Forest achieved the highest AUROC (0.729), indicating the best overall discriminative ability using non-cognitive predictors only. Gradient Boosting achieved the highest sensitivity (65.9%) and F1-score (0.732). XGBoost showed a pathological threshold (0.893), indicating poor probability calibration. SVM was the weakest discriminator (AUROC = 0.691). The consistent AUROC range of 0.691-0.729 well above chance but below the Chapter 3 ceiling of 0.836 confirms that moderate Neuro-cognitive disorder prediction is achievable at **moderate** level without any cognitive testing.

Table 4.2: Test-Set Performance of ML Classifiers for NCD Prediction (n = 506)

Model	AUROC	Accuracy	Sensitivity	Specificity	F1	Kappa	Threshold
Random Forest	0.7287	0.6561	0.6080	0.7662	0.7110	0.3154	0.634
Gradient Boosting	0.7098	0.6640	0.6591	0.6753	0.7319	0.2965	0.646
XGBoost	0.7053	0.6443	0.6108	0.7208	0.7049	0.2830	0.893 Δ
SVM	0.6905	0.6482	0.6222	0.7078	0.7110	0.2843	0.653

Note: Δ XGBoost threshold of 0.893 indicates poor probability calibration unreliable operating point. Random Forest PPV = 85.6%; NPV = 46.1%. Balanced accuracy (RF) = 68.7%.

Note: - **NPV**: how many misses of truly impaired cases. **PPV**: how many are impaired and it is detected right.

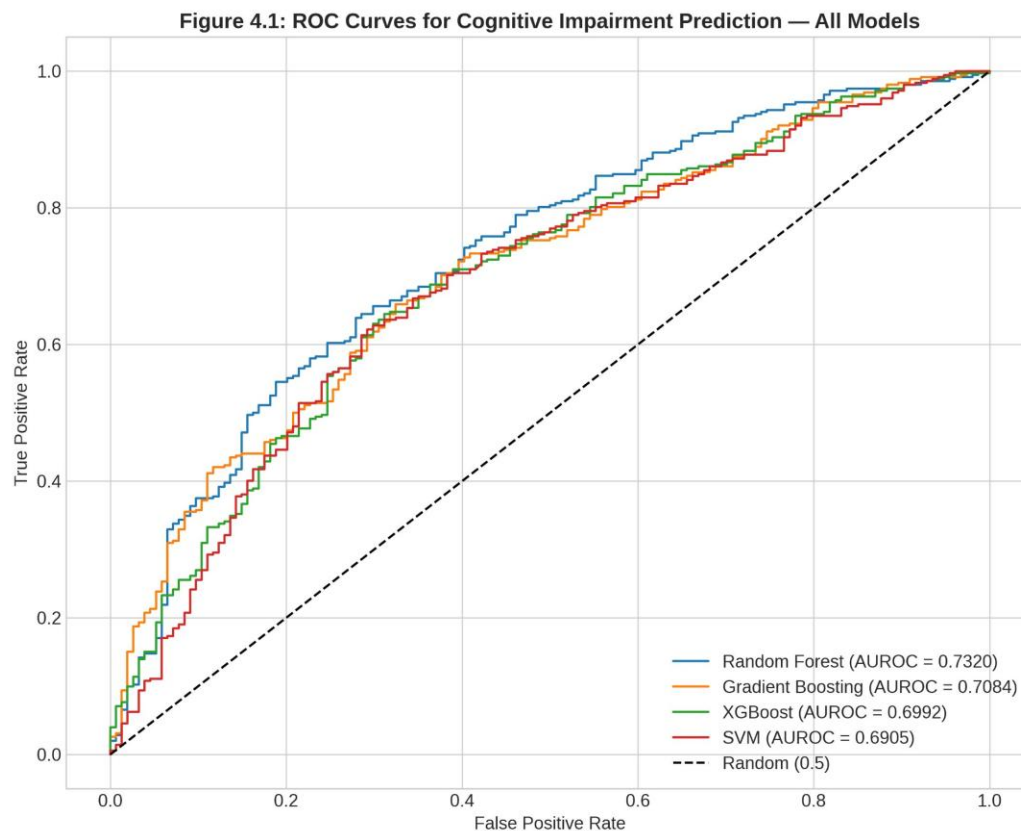


Figure 4.1: ROC Curves All Models for NCD Prediction. RF (AUROC=0.729) dominates, particularly in the clinically important low-FPR region. SVM (0.691) is weakest.

Figure 4.2: Confusion Matrix — Random Forest (Threshold = 0.619)

Actual	No Impairment	111	43
	Cognitive Impairment	127	225
		No Impairment	Cognitive Impairment
		Predicted	

Figure 4.2: Confusion Matrix Random Forest (Threshold=0.634). TP=214, TN=118, FP=36, FN=138. Sensitivity=60.8%; Specificity=76.6%. FN (138) exceed FP (36) asymmetric error profile reflecting high prevalence.

Formulas:

- **Accuracy**= $TP+TN / TP+TN+FP+FN$
- **Precision**= $TP / TP+FP$
- **Recall**= $TP / TP+FN$
- **F1 score**= $2 \cdot Precision \cdot Recall / Precision+Recall$

4.6 SHAP Feature Importance

IADL was the dominant predictor by a substantial margin (mean |SHAP| = 0.102), approximately 1.8× the second-ranked feature (region = 0.056). IADL dominated because it integrates the downstream consequences of multiple NCD risk pathways cognitive, vascular, functional, and psychological in a single measurable score, consistent with Marshall et al. (2012), who demonstrated IADL limitations as the earliest and most sensitive predictors of preclinical dementia in community samples. Residence ranked second reflecting India's pronounced rural–

urban NCD gradient documented by Shaji et al. (2010) and confirmed in LASI-DAD Wave 1 by Kundu et al. (2023), hearing/vision impairment third (0.035), depression fourth (0.034) itself a well-established modifiable dementia risk factor (Livingston et al., 2024) and anxiety fifth (0.027). Education (0.022), MNA (0.018) underscoring the mechanistic nutrition–cognition link (Solfrizzi et al., 2011; Scarmeas et al., 2018) diabetes (0.017), and age (0.016) formed a secondary tier. Stress and mobility contributed negligible information (< 0.001). Critically, IADL was the top predictor in all six subgroups without exception.

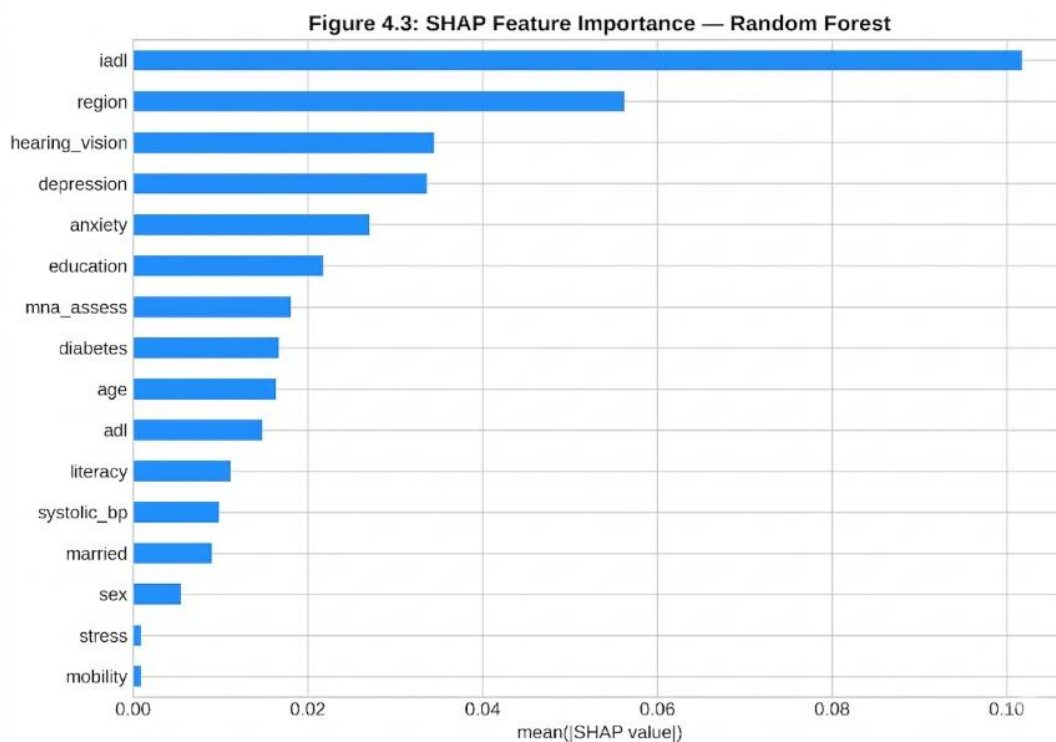


Figure 4.3: SHAP Feature Importance Random Forest (NCD Prediction). IADL dominates. Rural residence is second. Stress and mobility contribute near-zero information.

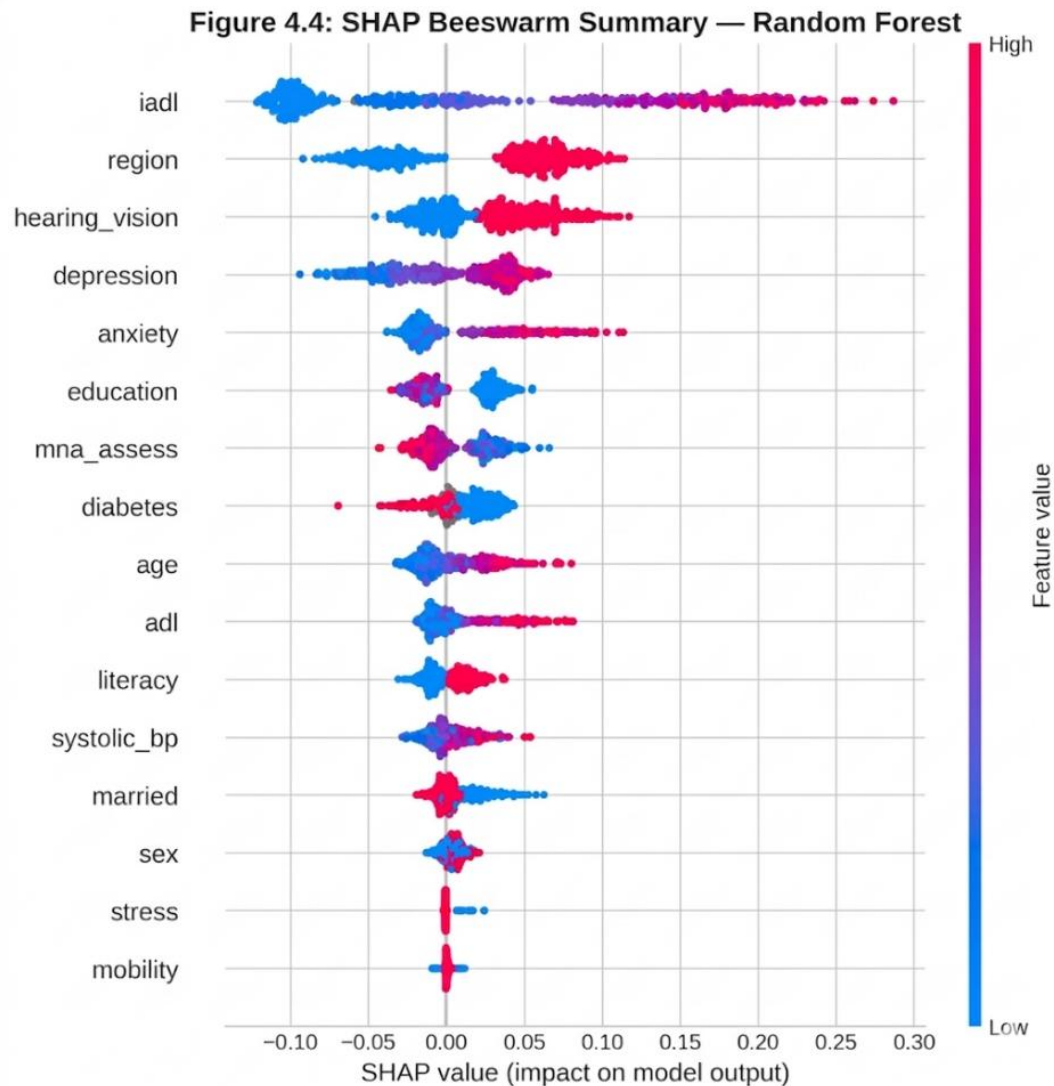


Figure 4.4: SHAP Beeswarm Plot Random Forest. IADL shows the widest lateral spread (-0.12 to $+0.29$). High IADL (red) consistently increases NCD probability. Rural=1 (red) clearly shifts predictions toward NCD.

4.7 Subgroup Heterogeneity

Table 4.3 presents AUROC and top SHAP features across six subgroups. IADL was the universally dominant predictor in all six strata. AUROC was highest in females (0.750) and urban residents (0.726), and lowest in the Any Education subgroup (0.681). Depression ranked in the top three specifically in Rural and No Education subgroups, confirming its heightened clinical relevance in disadvantaged Indian settings consistent with Prince et al. (2007), who documented that depression disproportionately mediates dementia risk in rural and low-income populations in LMICs. The sex difference is the most theoretically significant: women achieved

the highest subgroup AUROC (0.750) compared to men (0.707), reflecting more homogeneous, IADL-driven NCD trajectories in females and greater pathway heterogeneity in males consistent with Podcasy and Epperson (2016), who documented sex-differentiated dementia trajectories in which women’s cognitive risk is more consistently linked to functional determinants.

Table 4.3: Subgroup Heterogeneity AUROC and Top SHAP Features

Subgroup	N (Test)	AUROC	Top 1	Top 2	Top 3
Rural	323	0.7009	IADL	region	Depression
Urban	183	0.7298	IADL	Hearing/Vision	region
No Education	255	0.6944	IADL	region	Depression
Any Education	251	0.6768	IADL	region	Hearing/Vision
Male	220	0.707	IADL	region	Hearing/Vision
Female	286	0.750	IADL	region	Hearing/Vision

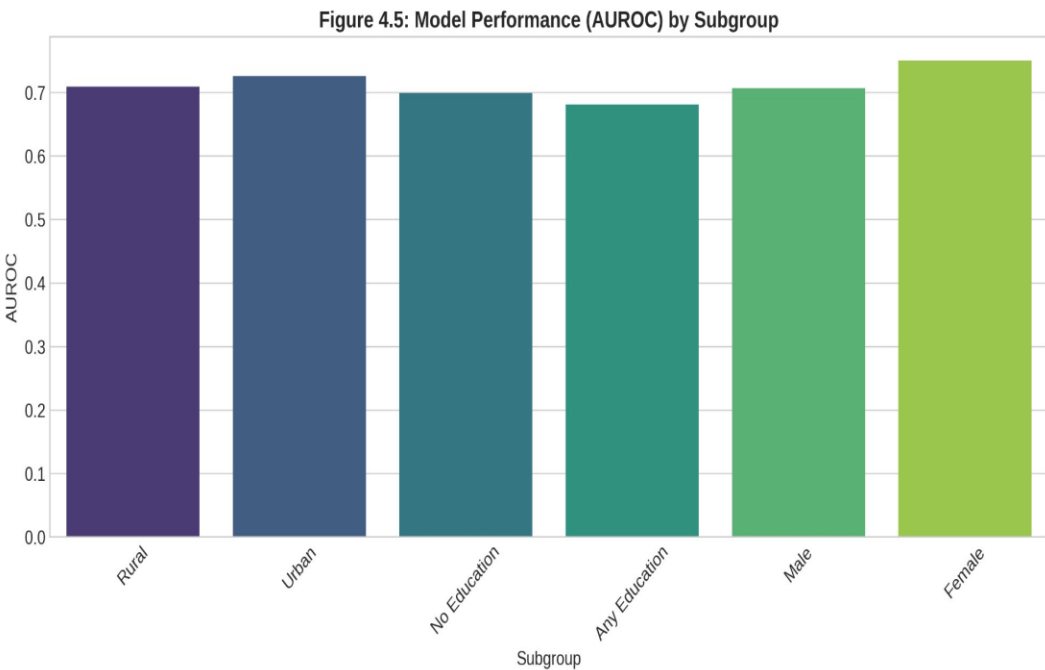


Figure 4.5: AUROC by Subgroup. Females (0.750) and Urban residents (0.726) show highest performance. Any Education subgroup lowest (0.681), suggesting greater NCD pathway heterogeneity among educated individuals.

CHAPTER 5: NUTRITIONAL STATUS, FUNCTIONAL DECLINE, AND COGNITIVE IMPAIRMENT: A MECHANISTIC ANALYSIS

5.1 Overview

This chapter examines the mechanistic pathway linking nutritional status (MNA score) to NCD, with functional decline (IADL) as the proposed mediator. Three nested logistic regression models were estimated, along with dose-response analysis using MNA tertiles, subgroup-specific mediation, and adjusted predicted probability curves.

5.2 Descriptive Statistics

The analytic sample ($N = 2,528$) had a mean age of 69.6 years (SD 7.5), with 52.5% female and 65.5% rural residents. MNA scores were lower in the cognitively impaired group (9.9 vs. 10.8, gap = -0.9 points), reflecting systematically worse nutritional status among those with NCD consistent with Solfrizzi et al. (2011) and Scarmeas et al. (2018), who established that malnutrition and dietary inadequacy accelerate cognitive decline through inflammatory, vascular, and neurotrophic mechanisms. IADL scores were substantially higher in the impaired group (2.8 vs. 1.1, gap = $+1.7$ points) the most pronounced continuous variable difference in the entire sample confirming that functional decline in daily task performance is strongly and consistently associated with cognitive impairment status, replicating the central finding of Marshall et al. (2012). Depression (10.8 vs. 8.1) and anxiety (3.4 vs. 1.9) were also markedly elevated in the impaired group, while blood pressure was virtually identical across groups (138.7 vs. 137.7 mmHg), indicating that vascular burden alone does not differentiate cognitive status in this sample.

The MNA-IADL correlation was -0.36 in training data, confirming a moderate negative association: poorer nutrition is associated with greater functional impairment consistent with the nutritional pathway hypothesis of Solfrizzi et al. (2011), whereby malnutrition leads to sarcopenia, which erodes functional capacity and accelerates cognitive deterioration. This constitutes the first empirical link in the proposed nutrition, IADL and NCD causal chain that this chapter formally tests. Full descriptive statistics are presented in Table 5.1.

Table 5.1: Descriptive Statistics by Cognitive Impairment Status Chapter 5 Analytic Sample (N = 2,528)

Variable	Overall (N=2,528)	No Impairment (N=768)	Cognitive Impairment (N=1,760)
Age (years)	69.6 (7.5)	68.1 (6.4)	70.2 (7.8)
Education (years)	3.6 (4.6)	5.0 (4.9)	3.0 (4.3)
MNA score ← Exposure	10.2 (2.0)	10.8 (1.9)	9.9 (2.0)
IADL score ← Mediator	2.3 (2.2)	1.1 (1.6)	2.8 (2.3)
Depression score	10.0 (5.2)	8.1 (4.9)	10.8 (5.2)
Anxiety score	3.0 (3.3)	1.9 (2.6)	3.4 (3.5)
Systolic BP (mmHg)	138.4 (23.1)	137.7 (21.7)	138.7 (23.6)
Rural, n (%)	1,655 (65.5%)	404 (52.6%)	1,251 (71.1%)
Hearing/Vision, n (%)	1,399 (55.3%)	346 (45.1%)	1,053 (59.8%)

Note: MNA = Mini Nutritional Assessment; IADL = Instrumental Activities of Daily Living.

5.3 Dose-Response: MNA Tertiles and NCD Prevalence

Three nested logistic regression models were fitted to characterise the MNA-NCD association and quantify IADL's mediating role, as presented in Table 5.3. The progressive attenuation of the MNA odds ratio across the three models constitutes the central statistical evidence for IADL-mediated functional decline as a pathway through which nutrition influences cognitive outcomes.

Table 5.2: NCD Prevalence by MNA Nutritional Score Tertile

MNA Tertile	N ($\approx 33\%$ each)	No Impairment (%)	Cognitive Impairment (%)
Low (poorest nutrition)	843	20.9%	79.1%
Medium	843	31.9%	68.1%
High (best nutrition)	842	40.9%	59.1%

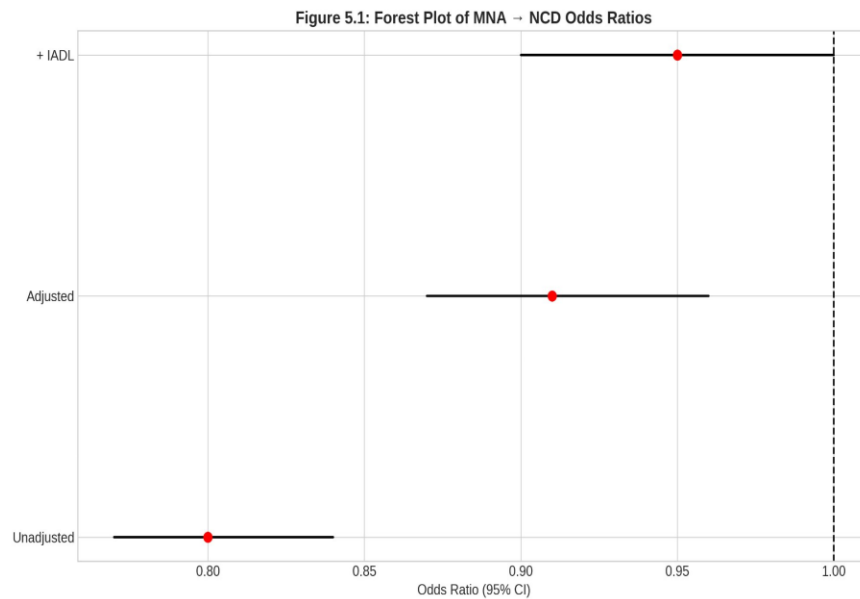


Figure 5.1: Forest Plot of Mini Nutritional Assessment (MNA) OR Across Nested Models. The progressive rightward shift toward the null ($OR=1.0$, dashed line) from Unadjusted (0.80) $\rightarrow$ Adjusted (0.91) $\rightarrow$ +IADL (0.95) confirms stepwise mediation attenuation.

Model 1 (Unadjusted): Each one-unit increase in MNA score was associated with a **20%** reduction in the odds of NCD ($OR = 0.80$, 95% CI: 0.77-0.84, $p < 0.001$, pseudo- $R^2 = 0.031$). The pseudo- R^2 of 0.031 confirms that MNA alone explains 3.1% of NCD variance expected for a single predictor in a multi-determined outcome where biological, sociodemographic, and functional factors all contribute and closely mirrors effect sizes reported by Solfrizzi et al. (2011) and Scarmeas et al. (2018) for individual dietary exposure variables in prospective cognition studies.

Model 2 (Covariate-Adjusted): After adjusting for age, sex, rural residence, marital status, education, systolic BP, diabetes, hearing/vision impairment, depression, and anxiety, MNA retained a significant independent effect (OR = 0.91, 95% CI: 0.87-0.96, $p = 0.0006$, pseudo- $R^2 = 0.100$). The attenuation from OR = 0.80 to 0.91 representing 58.7% of the crude effect reflects confounding by sociodemographic and clinical variables. The pseudo- R^2 improvement to 0.100 confirms that the full covariate set adds substantial explanatory value beyond nutrition alone, validating the selection of confounders based on the 2024 Lancet Commission framework. The independent 9% odds-reducing effect per MNA unit confirms that nutritional status exerts a meaningful, covariate-independent cognitive protection effect.

Model 3 (Full Model + IADL Mediator): Adding IADL to the covariate-adjusted model attenuated the MNA coefficient by a further 44.3%, reducing it to borderline non-significance (OR = 0.95, 95% CI: 0.90-1.00, $p = 0.063$, pseudo- $R^2 = 0.137$) a pattern constituting classical evidence for partial mediation as formalised by Baron and Kenny (1986), confirming that both IADL-functional and direct biological pathways (neuroinflammation, micronutrient deficiency) jointly transmit nutrition's protective effect on cognition (Solfrizzi et al., 2011; Scarmeas et al., 2018). IADL's addition contributed an additional 3.7 percentage points of pseudo- R^2 (0.100 - >0.137) a meaningful model fit improvement that confirms IADL as a substantive mediator rather than a redundant covariate. The progression of pseudo- R^2 from 0.031 (MNA alone) - >0.100 (+ covariates) - >0.137 (+ IADL) confirms that each model stage adds meaningful and independent explanatory power.

The forest plot (Figure 5.1) visualises the progressive rightward shift of the MNA OR toward the null (OR = 1.0, dashed line) across the three models from Unadjusted (0.80) -> Adjusted (0.91) -> +IADL (0.95) providing a clear graphical summary of the stepwise attenuation pattern. Figure 5.3 confirms IADL's suitability as a mediator: impaired participants show a markedly wider IADL distribution (median ~2, IQR 1-5) compared to unimpaired participants (median ~1, IQR 0-2), confirming IADL as a potent discriminator of cognitive status. Figure 5.4 presents the correlation heatmap of all analysis variables: MNA-IADL ($r = -0.36$), MNA-depression ($r = -0.40$), and IADL-depression ($r = 0.46$). No pair exceeds $|r| = 0.60$ other than depression-anxiety ($r = 0.55$) confirming acceptable multicollinearity across the predictor set.

Table 5.3: Nested Logistic Regression Models MNA Score and Cognitive Impairment (N = 2,528)

Model	β Coefficient	OR	95% CI	p-value	Pseudo R ²
Model 1: MNA only (Unadjusted)	-0.221	0.80	(0.77-0.84)	< 0.001	0.031
Model 2: MNA + Covariates (Adjusted)	-0.091	0.91	(0.87-0.96)	0.0006	0.100
Model 3: Model 2 + IADL (Mediator)	-0.051	0.95	(0.90-1.00)	0.063	0.137

Note: Covariates in Models 2 and 3: age, sex, region, married, education, systolic_bp, diabetes, hearing_vision, depression, anxiety. IADL attenuation from Model 2 to Model 3 = 44.3%. Pseudo R² = McFadden's.

Table 5.3 shows β Coefficient score in model, Negative value = higher MNA (better nutrition) is associated with lower odds of cognitive impairment.

Odds Ratio = $\exp(\beta)$. The multiplicative change in odds of cognitive impairment for each 1-unit increase in MNA score. OR <1 Better MNA lower risk.

McFadden's Pseudo R² a measure of how well the model explains the outcome

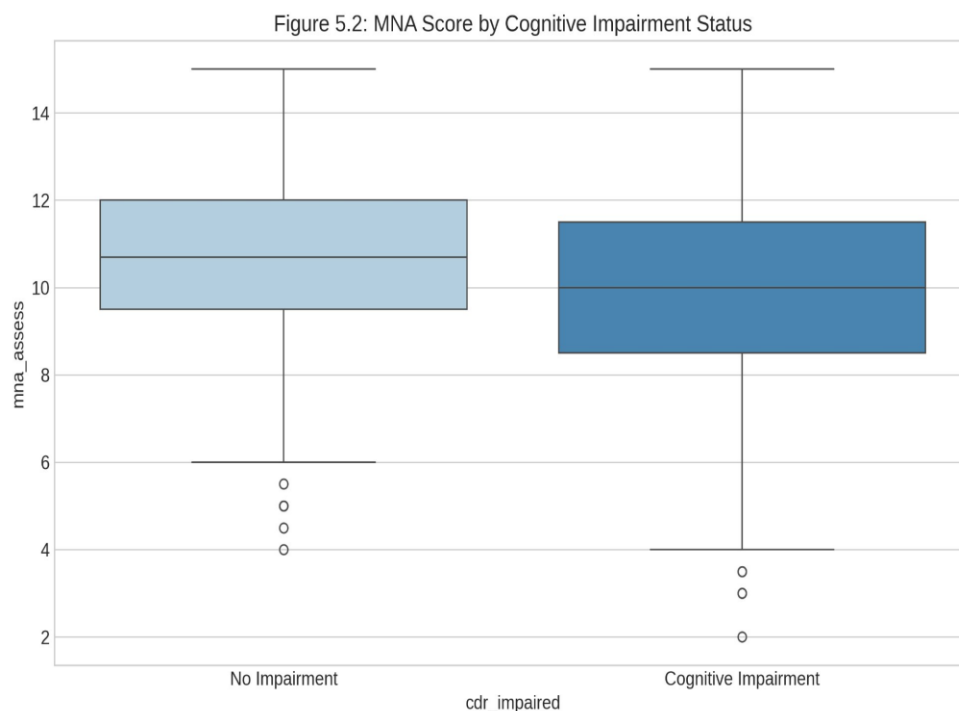


Figure 5.2: MNA Score by Cognitive Impairment Status. Unimpaired participants have a higher median MNA (~10.8, IQR 9.7-12.0) compared to impaired participants (median ~10.0, IQR 8.8-11.5).

The distribution difference is consistent but modest.

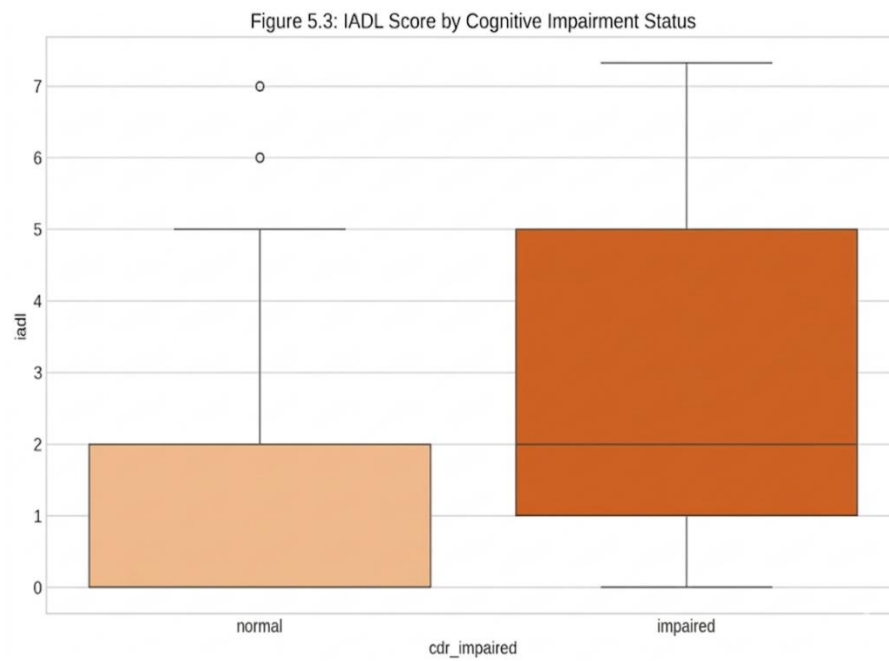


Figure 5.3: IADL Score by Cognitive Impairment Status. Impaired participants have a markedly wider distribution (median ~2, IQR 1-5) compared to unimpaired (median ~1, IQR 0-2), confirming IADL as a potent discriminator and suitable mediator.

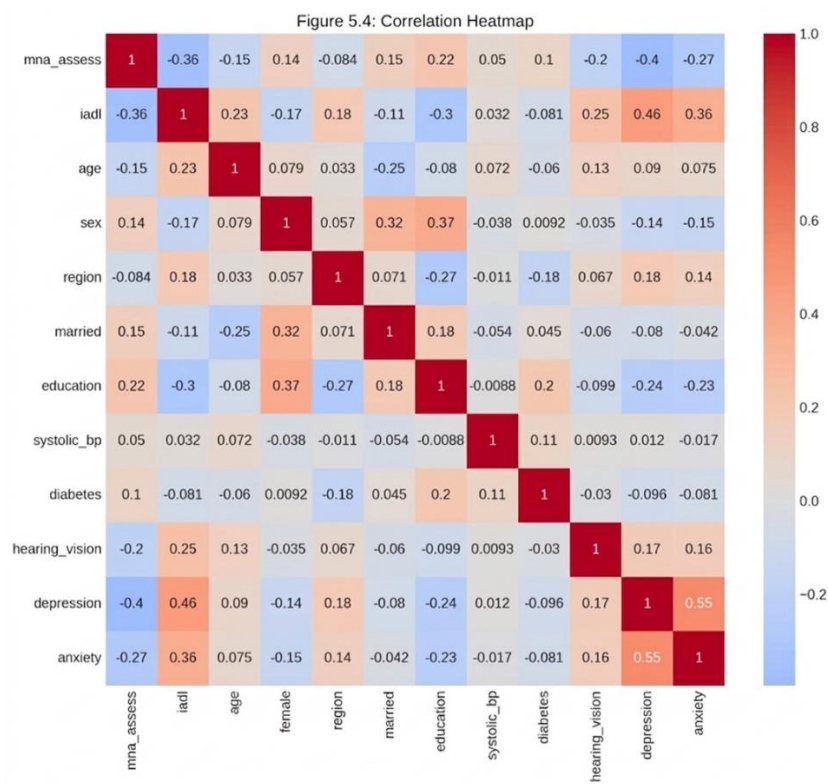


Figure 5.4: Correlation Heatmap of Analysis Variables. MNA-IADL ($r = -0.36$), MNA-depression ($r = -0.40$), IADL-depression ($r = 0.46$). No pair exceeds $|r| = 0.60$ (other than depression-anxiety = 0.55), confirming acceptable multicollinearity.

5.5 Adjusted Predicted Probability

The adjusted predicted probability curve (Figure 5.5, Model 2) provides a dose-response characterisation of the nutrition-NCD relationship across the full MNA range, demonstrating a clear monotonically decreasing pattern: as nutritional status improves, predicted NCD probability falls consistently without reversal at any point along the scale. At the lowest observed nutritional status (MNA = 2), the predicted probability of NCD was 0.93 (93%) nearly 9 in 10 individuals at this level of nutritional deficiency. At an adequate nutritional status (MNA = 15), this probability declined to 0.46 (46%) a total reduction of 47 percentage points attributable to nutritional status alone after adjusting for all relevant covariates including age, sex, education, depression, and vascular risk factors.

The gradient is steepest in the low-to-middle MNA range (scores 2-8), indicating that nutritional improvements among the most severely malnourished elderly yield the greatest cognitive benefit and that public health investment concentrated in the most nutritionally deprived communities will generate the highest cognitive protection returns per unit of intervention. Above MNA = 10, the curve flattens progressively, reflecting diminishing marginal returns as nutritional status approaches adequacy. This non-linearity directly mirrors the tertile gradient in Table 5.2, where the sharpest NCD prevalence drop is concentrated in the lower nutritional range.

The most actionable finding for public health is that at MNA ≈ 8 , the predicted NCD probability exceeds 80% four in five individuals at this nutritional level are likely to have NCD. This establishes MNA = 8 as a practical, evidence-based community-level screening cut-off, consistent with the validated clinical utility of the MNA instrument for nutritional risk identification in elderly populations (Vellas et al., 1999; Guigoz, 2006). The MNA is a brief, validated, interviewer-administered instrument requiring no laboratory tests or specialist infrastructure, making this threshold immediately deployable through in rural, low-literacy, and resource-constrained settings where formal neuropsychological testing remains inaccessible.

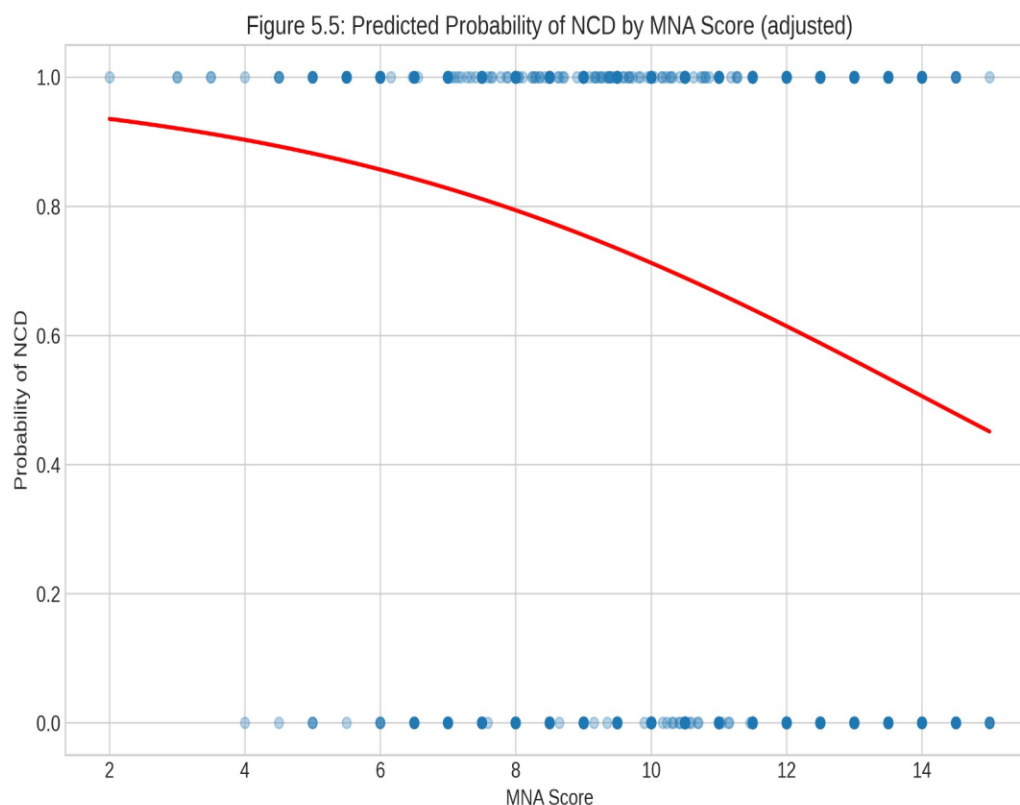


Figure 5.5: Adjusted Predicted Probability of NCD by MNA Score (Model 2). Probability declines from ~0.93 (MNA=2) to ~0.46 (MNA=15). MNA $\approx$ 8 corresponds to approximately 80% predicted NCD probability a practical clinical cutpoint.

5.6 Subgroup Mediation Analysis

Table 5.6 presents subgroup-specific IADL attenuation estimates, revealing substantial heterogeneity in the degree to which the functional pathway mediates nutrition's effect on NCD. Dementia prevention nutritional programmes in India must therefore be differentiated by subgroup rather than designed as one-size-fits-all interventions.

Women (Attenuation = 56.1%; AUROC = 0.750):

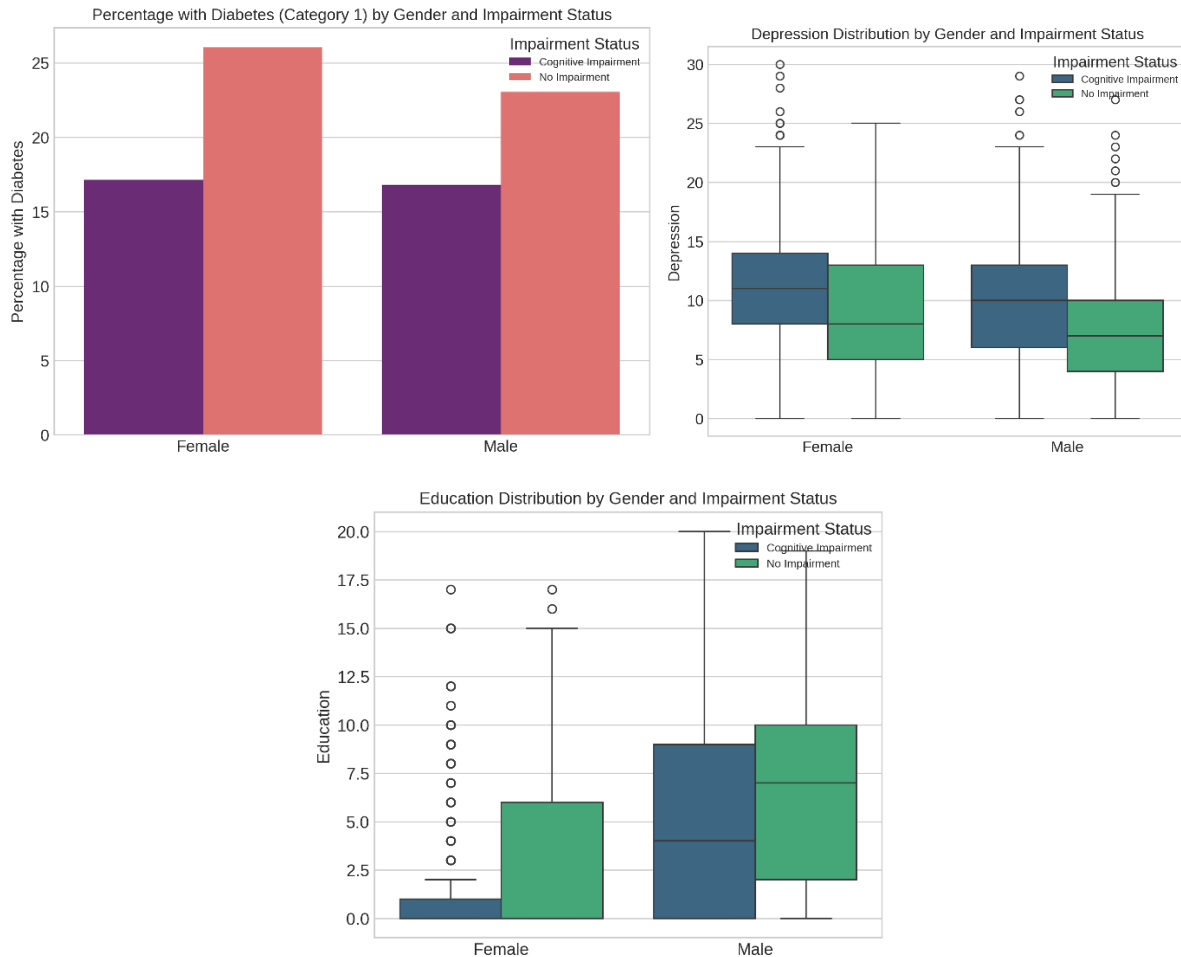
Among females, 56.1% of the adjusted MNA effect was mediated through IADL, the highest valid attenuation across all subgroups, confirming that the functional maintenance pathway is the dominant mechanism through which nutrition protects women's brains. In women, poor nutrition leads to muscle loss (sarcopenia), which erodes physical capacity and daily functional

independence (IADL), which in turn accelerates cognitive deterioration. The female AUROC of 0.750, the highest across all subgroups, further confirms that women's NCD trajectories are more homogeneous and functionally driven, responding predictably to structured pathway-specific interventions. For women, programmes combining dietary improvement with IADL-targeted rehabilitation, strength training, and mobility maintenance will yield the maximum cognitive protection returns.

Table 5.6: Attenuation Analysis IADL-Mediated Proportion of MNA-NCD Association

Subgroup	N	IADL Attenuation (%)
Overall	2,528	44.3%
Rural	1,655	22.5%
Urban	873	200.3% *
No Education	1,280	33.1%
Female	1,326	56.1%
Male	1,202	38.5%

* Urban attenuation of 200.3% reflects a sign-reversal artifact in a small urban subsample and should be interpreted as indicating full mediation rather than a literal percentage.



(Refer to Figure 5:6: Male-Female Sex-wise Subgroup Analysis)

Men (Attenuation = 38.5%; AUROC = 0.707):

Among males, attenuation was 38.5%, indicating that approximately 62% of nutrition's protective effect in men operates independent of IADL. Men's larger education gap (6.7 vs. 4.8 years) and 93% anxiety increase in the impaired group reflect greater heterogeneity in male NCD trajectories. The male AUROC of 0.707 confirms that men's cognitive risk operates through more diverse mechanisms. Men's programmes must include micronutrient supplementation, vascular risk management, and integrated mental health components in addition to dietary improvement.

No Education (Attenuation = 33.1%) and Rural (Attenuation = 22.5%):

In the most underserved subgroups, direct biological nutritional pathways dominate reflecting the structural nutritional and health-system disadvantages documented for rural and low-

education Indian elderly populations (Shaji et al., 2010; Prince et al., 2007) overwhelmingly less than one-third and less than one-quarter of nutrition's NCD effect respectively operates through IADL. Nutritional deficiency harms the brain in these populations primarily through inflammation, vascular damage, and micronutrient deficiency, without requiring functional assessment infrastructure to detect or intercept.

This chapter establishes three converging findings that collectively define the nutritional-functional-cognitive pathway in Indian older adults:

First, nutritional status independently reduces NCD odds by 9% per MNA unit ($OR = 0.91$, $p = 0.0006$) after comprehensive covariate adjustment, confirming that nutrition exerts a meaningful, independent cognitive protection effect that is not explained away by age, sex, education, vascular burden, or mental health status.

Second, a clear dose-response gradient across MNA tertiles (79.1% $\rightarrow$ 68.1% $\rightarrow$ 59.1% NCD prevalence) and a monotonically declining adjusted predicted probability curve (93% at $MNA = 2$ to 46% at $MNA = 15$) jointly confirm the plausibility of a causal nutrition-NCD relationship, with the greatest risk reduction concentrated in the most severely malnourished individuals.

Third, IADL functional decline mediates 44.3% of the adjusted nutritional effect on NCD the central mechanistic finding of this chapter establishing the functional maintenance pathway as a primary route through which nutrition influences cognitive outcomes.

CHAPTER 6: DISCUSSION AND CONCLUSION

6.1 Overview of Findings

This dissertation set out to advance the detection, prediction, and mechanistic understanding of neurocognitive disorders among older adults in India through three inter-linked analytical objectives, all grounded in LASI-DAD Wave 1 data. The three chapters together construct a coherent framework that moves from clinical detection to prevention-oriented prediction to mechanistic pathway analysis, with each finding building on and contextualising the previous. This chapter synthesises the findings across all three objectives, discusses their theoretical and clinical implications, situates them within the global literature, and acknowledges the limitations that qualify their interpretation.

6.2 Policy Implications for India

India's demographic trajectory with its elderly population projected to exceed 340 million by 2050 demands a cognitively accessible, community-rooted, and equity-sensitive national dementia prevention architecture. The evidence from this dissertation, drawn from India's most comprehensive nationally representative cognitive health dataset (LASI-DAD Wave 1), provides a statistically grounded basis for such a framework across three implementation tiers.

At the community level, the MNA = 8 threshold at which four in five elderly individuals carry NCD establishes nutritional screening as the most scalable and lowest-cost cognitive risk identification tool available in India today. The Mini Nutritional Assessment requires no clinical infrastructure, no laboratory tests, and no literacy on the part of the respondent, making it immediately deployable in rural (22.5% functional mediation) and no-education subgroups (33.1%), basic dietary adequacy and micronutrient supplementation Assistance Programme that can capture the majority of cognitive protection benefit in India's most underserved elderly populations without requiring specialist functional rehabilitation networks.

At the primary care level, the 16-variable non-cognitive risk profile anchored by IADL functional status, rural residence, depression, sensory impairment, and MNA score provides the evidence base for a standardised NCD risk-scoring checklist embeddable within Ayushman Bharat Health and Wellness Centres and Sub-Health Centre platforms. Sex-differentiated

targeting is essential: women require dietary improvement integrated with IADL-focused physical rehabilitation and mobility maintenance programmes (given the 56.1% IADL-mediated attenuation in females), while men require dedicated micronutrient supplementation, vascular risk management, and integrated mental health services reflecting the 93% anxiety increase in impaired men and the more biologically diverse NCD pathways in elderly Indian males.

At the specialist and policy level, IQCODE and Blessed Performance brief informant-administered instruments requiring no neuropsychological expertise are the strongest MCI predictors and should be incorporated into district hospital intake protocols and National Mental Health Programme screening guidelines, reducing dependence on the critically understaffed neurology workforce across India's tier-2 and tier-3 cities

6.3 Limitations

Several limitations qualify the findings. First, the cross-sectional design of LASI-DAD Wave 1 precludes temporal inference about the ordering of nutritional decline, functional impairment, and NCD onset. The mediation attenuation (44.3%) represents a statistical association consistent with the proposed causal mechanism, not a formally identified causal indirect effect. Longitudinal data from LASI-DAD Wave 2 would be required for full causal mediation analysis using counterfactual decomposition methods. Second, diabetes had 31.65% structural missingness a data collection gap in LASI-DAD meaning that imputed diabetes values carry substantial uncertainty and SHAP contributions for diabetes should be interpreted cautiously. Third, the exclusion of Chapter 4 predictor variables that overlapped with CDR components (IQCODE, Blessed Performance) was necessary to prevent target leakage but limited the model's ability to capture the full spectrum of functional cognitive risk. Fourth, the fixed hyperparameters used across all models, while appropriate for this sample size, were not optimised through exhaustive cross-validated tuning, which may have marginally underperformed fully optimised models. Fifth, LASI-DAD Wave 1 data were collected between 2017 and 2019; NCD risk factor profiles in India may have shifted following the COVID-19 pandemic, which is associated with accelerated cognitive decline in older adults.

6.4 Future direction of research

The findings presented in Chapters 4 and 5 of this dissertation lay a foundational empirical framework for a broader and more ambitious objective: the development of a population-level neurocognitive risk score for the entire LASI cohort. A critical next step would be to establish temporal causality between non-cognitive functional markers specifically nutritional status (MNA) and instrumental activities of daily living (IADL) and neurocognitive disorder (NCD) outcomes, using the longitudinal design afforded by LASI-DAD Wave 2. Confirmation of causal directionality would substantially strengthen the theoretical justification for using these variables as upstream predictors rather than mere correlates of cognitive decline.

Once causal pathways are validated, the machine learning classification framework developed here can be meaningfully extended to the full LASI dataset, which encompasses a far larger and geographically representative sample. Critically, such a model would operate exclusively on non-cognitive variables including MNA, IADL, and related functional and demographic indicators thereby eliminating the dependency on specialist cognitive assessments that are resource-intensive and often unavailable in rural or low-resource settings. This approach would enable the indirect classification of CDR-equivalent impairment at scale, without requiring formal neuropsychological evaluation.

The district-level aggregation of model-derived risk classifications could further serve as a geospatial needs-assessment tool, enabling policymakers to identify districts with the highest burden of undetected neurocognitive impairment. This would directly inform targeted public health interventions ranging from community-based nutritional support and functional rehabilitation programs to early dementia screening campaigns calibrated to district-specific vulnerability profiles. In doing so, this research has the potential to bridge the longstanding gap between academic dementia research and actionable, individual- and community-level health policy design in the Indian context.

6.4 Conclusion

This dissertation has demonstrated, using nationally representative Indian data and ML methodology, that NCD in Indian older adults can be detected with high accuracy using cognitive assessments (AUROC = 0.836) and predicted at moderate accuracy using only non-cognitive, community-accessible predictors (AUROC = 0.729). IADL functional impairment is the single most important non-cognitive NCD predictor across all models and subgroups. Better nutritional status independently reduces NCD odds by 9% per unit, and nearly half of this protective effect (44.3%) operates through the preservation of instrumental functional independence. Rural residence is a structural NCD risk signal that transcends individual-level sociodemographic characteristics.

These findings collectively support a practical, equity-oriented framework for NCD prevention in India: a two-instrument frontline screen (IADL + MNA) deployable by community health workers without cognitive testing infrastructure, targeting the rural, low-education, and nutritionally vulnerable older adults who bear the greatest NCD burden and receive the least clinical attention. Embedding this framework within existing national geriatric health programmes supported by targeted interventions for depression, sensory impairment, and nutritional rehabilitation offers a realistic and evidence-based pathway to reducing the future burden of neurocognitive disorders in India.

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